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A Trajectory Planning Approach Incorporating Load Transfer Dynamics and Energy-Based Rollover Risk for Autonomous Truck Platoon

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***Abstract*—During high-speed platooning, if the leading vehicle experiences sudden failure (e.g., tire blowout or braking loss), the following vehicles must execute simultaneous deceleration and obstacle avoidance within a minimal timeframe. This process induces significant load transfer, leading to drastic variations in vertical load distribution and tire cornering stiffness, thereby critically compromising lateral stability. Conventional planning methods, reliant on static stability boundaries and fixed rollover thresholds, fail to adapt to such dynamic perturbations, resulting in degraded control performance. This paper proposes an intelligent trajectory planning framework for emergency obstacle avoidance in heavy-duty truck platoons. The framework incorporates a physics-informed neural network to predict real-time load transfer effects on tire cornering stiffness, combined with an Energy-based Rollover Index (ERI) for dynamic stability boundary assessment. Innovatively integrating the nonlinear effects of braking-induced load transfer into coordinated trajectory-speed optimization, the method enhances platoon maneuverability while ensuring safety. Simulation results demonstrate that the proposed approach increases maximum safe lane-change speeds by 0.5%-4.3% while reducing false rollover alarms to below 2%. Co-simulation using MATLAB/Simulink and TruckSim verifies a 4.2% improvement in overall planning performance while maintaining dynamic safety standards.**

***Index Terms*—Autonomous heavy-duty trucks platoons, adaptive trajectory planning, load transfer modeling, stability estimation, rollover risk.**

I. Introduction

As an important component of intelligent transportation systems, heavy-duty truck platooning has attracted increasing attention for its potential to enhance road throughput, reduce fuel consumption, and realize cooperative control. During high-speed platoon operations, trucks usually maintain short inter-vehicle distances to improve convoy efficiency [1]. In such tightly coupled configurations, when the leading vehicle experiences sudden failures, such as a tire burst, an abrupt stop, or an electronic system fault, the following trucks must quickly perform combined braking and lane-change maneuvers to avoid collisions. This rapid-response process intensifies dynamic changes in vehicle states, causing pronounced load transfer effects that significantly alter vertical load distribution [2] and, consequently, affect tire lateral force characteristics and stability boundaries [3]. Therefore, a planning framework capable of dynamically adapting to load transfer-induced stability variations is crucial for ensuring both safety and operational efficiency in intelligent truck applications.

Over the past decade, trajectory planning methods for autonomous vehicles have evolved considerably. Early-stage approaches primarily focused on geometric feasibility, employing techniques such as sampling-based algorithms, graph search strategies, and curve fitting methods (e.g., sampling-based [4], Bé zier curves [5], B-splines [6]). This limitation is particularly pronounced in scenarios such as high-speed emergency braking, where dynamic load transfer can significantly alter the vehicle's dynamics, which a planning strategy based on geometric constraints cannot accommodate. In platooning scenarios, the dynamic changes caused by load transfer are more pronounced because of the close interaction between vehicles in the convoy. From a broader systems perspective, these tightly coupled interactions in truck platoons can be regarded as a specific instance of cooperative motion planning and control in networked autonomous systems, as systematically reviewed for aerial–ground teams in [7]. Within such closely coordinated formations, the longitudinal and lateral coupling between vehicles, together with pronounced dynamic load transfer, leads to substantial variations in stability and safety margins, particularly during emergency maneuvers such as hard braking or lane-change avoidance [8]. Consequently, planning strategies that rely on static assumptions and neglect dynamic load-transfer effects are no longer sufficient to ensure safety and feasibility in dense platoons. This motivates the development of trajectory planning methods that explicitly incorporate dynamic load transfer into the planning and optimization framework.

Research has therefore advanced from purely geometric planning toward dynamic-constraint methodologies that

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explicitly embed vehicle dynamics and stability limits into the planning process. Within this line of work, multiphase and multi-objective optimal control formulations for autonomous ground vehicles and multi-UGVs illustrate how dynamic feasibility, comfort, and safety can be jointly handled in a unified trajectory optimization framework across different maneuver phases and mission scenarios [9], [10]. However, other studies demonstrate that emergency braking of unloaded trucks can induce more than 50% load transfer to the rear axle, with a nonlinear dependence on braking intensity [2], which poses challenges for conventional static stability analysis. Current stability-boundary modeling mainly relies on two approaches: phase-plane analysis [11], which can accurately identify rollover thresholds but neglects longitudinal coupling, and optimal control methods [12], which often suffer from computational complexity in handling constraints. To alleviate these difficulties, some studies have adopted machine learning techniques to construct regression models of phase-plane stability regions that incorporate tire characteristics [13], though substantial training data requirements constrain their applicability. In parallel, deep learning-based trajectory planning and control frameworks for AGV parking maneuvers and mobile robots operating in unknown environments have shown that neural networks can be trained from pre-optimized trajectories and exploration data to generate real-time motion profiles under complex constraints [14], [15]. Unlike these end-to-end or policy-learning schemes, the present work employs physics-informed neural networks solely to model load-transfer-aware stability boundaries and energy-based rollover risk indices, while keeping the vehicle dynamics in explicit analytical form. The movable stability-region approach further enhances steering adaptability through dynamic boundary adjustment, yet still fails to account for dynamic load transfer [16]. In the field of rollover risk assessment, methodologies are evolving from static to dynamic paradigms: traditional load transfer ratio (LTR) metrics, which cannot detect tripped rollovers, have been refined through vertical-acceleration integration [17] and adaptive estimation [18], [19]. Certain studies leverage neural network-based rollover time predictors [20], achieving high accuracy but with limited real-time performance. Novel rollover indices integrated with model predictive control [21] exhibit parameter sensitivity and poor generalizability. Notably, energy-based approaches incorporating lateral kinetic energy and virtual gravity concepts, combined with the roll stability index (RSI) [22] and LuGre model [23], establish a theoretical framework for stability regions that provides a direct foundation for the present study.

This paper proposes an adaptive trajectory-speed co-optimization framework for heavy-duty truck platoons to address the limitations of static constraint-based planning under dynamic braking-induced load transfer. The framework enables real-time generation of stability boundaries and intelligent assessment of rollover tendencies by leveraging a physics-informed neural architecture to model load transfer effects and a cumulative roll energy metric for risk evaluation. This integration improves feasibility and safety in motion planning for autonomous heavy-duty trucks, including both single-vehicle and multi-vehicle platooning scenarios. The main contributions are summarized as follows:

1) An adaptive trajectory–speed co-optimization framework is proposed for autonomous heavy-duty truck platoons, which dynamically adjusts stability constraints in response to braking-induced load transfer. This framework integrates vehicle dynamics with real-time maneuver planning in a closed-loop fashion, ensuring coordinated motion across all vehicles in the platoon. By accounting for inter-vehicle dynamics and load transfer between lead and following trucks, the framework demonstrates strong adaptability and autonomous decision-making capability under time-varying operating conditions in platoon scenarios.
2) A generalized energy-based rollover risk indicator (ERI) is designed, which transforms traditional rule-based thresholding into a continuous energy accumulation metric. The ERI enables intelligent risk discrimination by quantifying rollover tendencies across diverse maneuvers and vehicle configurations, particularly in platoon formations, where cooperative vehicle movement significantly affects rollover risk. This provides a robust, extendable tool for onboard safety management systems, ensuring that all vehicles in the platoon are evaluated for dynamic stability in real time.
3) A physics-informed neural network modeling architecture is developed to automate the estimation of load transfer rates and tire cornering stiffness. This approach achieves real-time generation of stability boundaries with high physical interpretability and low computational cost, facilitating intelligent constraint updating in the context of platoon-wide dynamics.

It is worth emphasizing that although the proposed framework is demonstrated under a truck platoon scenario, the entire trajectory planning and stability-constrained optimization framework is fundamentally established at the single-vehicle level. The platoon case is constructed by incorporating inter-vehicle spacing and coordination constraints on top of the single-truck formulation. By removing these cooperative constraints, the proposed method directly degenerates into a trajectory planning scheme applicable to an individual heavy-duty truck, without modifying the vehicle dynamics, stability boundaries, or rollover risk evaluation models.

The remainder of this paper is organized as follows. Section II presents the overall framework and main processing flow of the proposed method. Section III details the trajectory planning strategy. Section IV describes the modeling of dynamic constraints and the mechanisms for stability evaluation. Section V discusses simulation results and comparative analyses. Section VI concludes the paper and outlines future research directions.

To improve the reliability and adaptability of lane-change planning for autonomous heavy-duty truck platoons, this study proposes a trajectory–speed co-optimization framework that integrates dynamic constraint learning with energy-based rollover risk evaluation. The overall architecture and data flow are illustrated in Fig. 1. The framework explicitly accounts for

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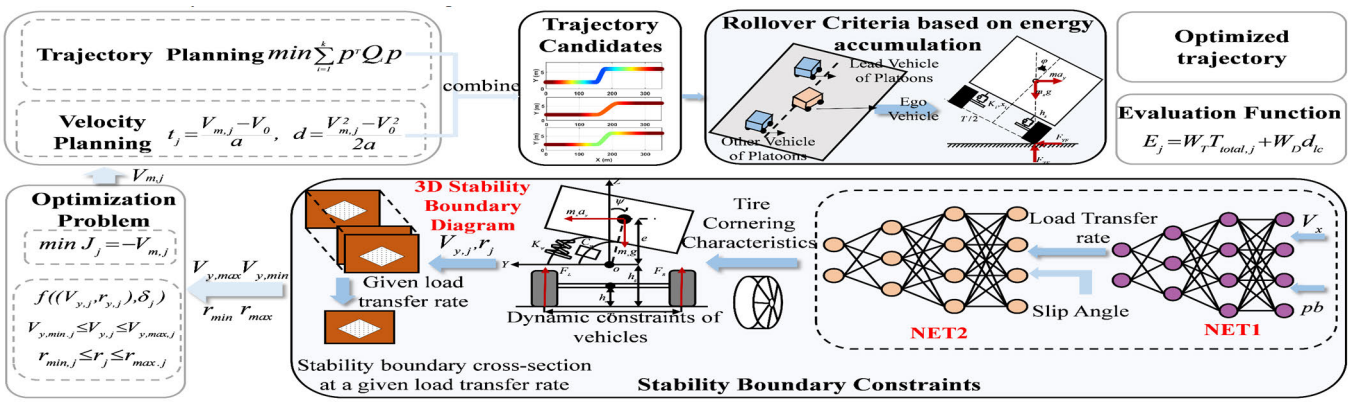

Fig. 1. Overall trajectory–speed cooperative planning framework considering braking-induced load transfer.

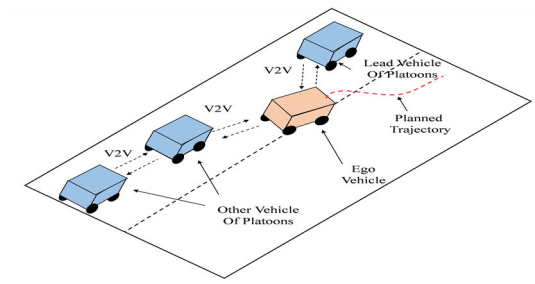

Fig. 2. Emergency obstacle avoidance scenario for truck platoon.

braking-induced load transfer in tightly spaced platoons, where close inter-vehicle interactions cause significant variations in vertical loads and tire cornering stiffness during high-speed maneuvers. A two-stage physics-informed neural network predicts load transfer rates and lateral force characteristics from the current braking state, initial vehicle velocity, and tire slip angles. These predictions are used to update the stability boundaries in real time, thereby capturing braking-adaptive stability variations within the platoon. Based on the dynamic stability boundaries, a constrained optimization module computes the maximum safe lane-change speed for the following truck and generates candidate trajectories that satisfy both kinematic and dynamic constraints. Each candidate is then evaluated using a cumulative roll energy metric to quantify the maneuver's rollover risk, and a multi-objective selection process finally chooses the optimal trajectory by jointly considering stability margins, rollover tendency, and maneuver efficiency within the cooperative control framework.

II. Trajectory Planning of Vehicle in The Platoons

The trajectory planning method proposed in this study determines the optimal lane-change trajectory through a three-phase optimization process: "braking deceleration, lane-change execution, and acceleration recovery" [24]. When the vehicles in the platoon receive the fault information from the preceding vehicle via V2V communication, they immediately adopt the emergency obstacle-avoidance strategy, as illustrated in Figure 2.

A. Trajectory Planning Based on Minimum Snap

The lane-change trajectory planning in this study employs the Minimum Snap method ([25], [26]) to solve the quadratic programming problem Equation (2) for obtaining x- and y-axis solutions, thereby constructing a complete candidate trajectory set. Based on the vehicle's geometric features and the obstacles, collision detection eliminates infeasible paths, ensuring that all candidate trajectories satisfy collision-free constraints and continuity requirements for tracking control. An offline pre-computation and online table-lookup strategy is adopted to enhance real-time performance. The trajectory can be expressed in vector form as Equation (1):

$$p(t) = [1, t, t^2, \cdots, t^n] \cdot p \tag{1}$$

In the trajectory planning process employed in this study, multiple sampling points are selected, including k + 1 two-dimensional waypoints (comprising the start and end points $pt_0, pt_1, \ldots, pt_k$, $pt_i = (x_i, y_i)$. The initial and final velocities and accelerations are determined based on speed planning. The trajectory is divided into k segments according to these waypoints, with separate trajectory planning conducted for the x and y dimensions. The constructed Minimum Snap optimization function is formulated as follows:

$$min \sum_{i=1}^{k} \int_{t_{i-1}}^{t_i} \left(\left[0, 0, 0, 0, 24, \cdots, \frac{n!}{(n-4!)} t^{n-4} \right] \cdot p \right)^T \cdot \left[0, 0, 0, 0, 24, \cdots, \frac{n!}{(n-4!)} t^{n-4} \right] \cdot p \mathrm{d}t = \min \sum_{i=1}^{k} p^T Q_i p. \tag{2}$$

where,

$$Q_i = \begin{bmatrix} 0_{4\times4} & 0_{4\times(n-3)} \\ 0_{(n\text{-}3)\times4} & R \end{bmatrix} \tag{3}$$

$$R = \frac{r!}{(r-4)!} \frac{c!}{(c-4)!} \frac{1}{(r-4)+(c-4)+1} (t_i^{(r+c-7)} - t_{i-1}^{(r+c-7)}) \tag{4}$$

r and c represent the row and column indices of the matrix, respectively. At this stage, the single-dimensional trajectory planning problem is formulated as a quadratic programming (QP) problem.

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The corresponding constraints must be established after formulating the quadratic programming (QP) problem. First, equality constraints are imposed on the trajectory's position, velocity, and acceleration at each sampling point. Following the equality constraints, feasible corridor constraints are applied to each sampling point by adding position inequality constraints to restrict the trajectory shape. These constraints will not be further elaborated in this paper.

While the trajectory planning strategy in this paper primarily focuses on the single-vehicle approach, it is applied to heavy-duty truck platoons. In platooning, trucks operate with close inter-vehicle distances for improved efficiency, which introduces additional dynamic considerations, as the platoon scenario influences external constraints.

B. Velocity Planning

In the speed planning phase, this study focuses on safety and energy efficiency optimization tailored for highway scenarios, guided by three design criteria: (1) minimizing acceleration or deceleration frequency to improve ride comfort, (2) controlling load transfer magnitude to preserve tire cornering stiffness stability, and (3) maximizing travel efficiency under traffic regulation constraints. This design effectively balances the conflicting requirements between dynamic stability and motion smoothness.

This study employs a trapezoidal velocity planning strategy to achieve safe and efficient trajectory-tracking control [27]. The implementation consists of three key phases: (1) braking deceleration to the optimal lane-change velocity $V_{m,j}$; (2) maintaining constant velocity during the lane-change maneuver; (3) accelerating to restore the initial cruising speed. The determination of $V_{m,j}$ represents the core of velocity planning, with its acquisition method detailed in Section IV.

The corresponding trapezoidal velocity profile is established upon determining the optimal lane-change speed $V_{m,j}$. To simplify the configuration process, constant deceleration $a_{dec} < 0$ and acceleration $a_{acc} < 0$ are employed. Thus, to facilitate subsequent evaluation function design, the time durations and travel distances for both deceleration and acceleration phases are calculated as follows:

$$t_{dec,j} = \frac{V_{m,j} - V_0}{a_{dec}}, \quad d_{dec,j} = \frac{V_{m,j}^2 - V_0^2}{2a_{dec}}, \tag{5}$$

$$t_{acc,j} = \frac{V_0 - V_{m,j}}{a_{acc}}, \quad d_{acc,j} = \frac{V_0^2 - V_{m,j}^2}{2a_{acc}}. \tag{6}$$

The total time and total distance of the whole trajectory planning are as follows:

$$T_{sum,j} = t_{dec.j} + t_{m,j} + t_{acc,j}, \tag{7}$$

$$D_{sum,j} = d_{dec,j} + d_{m,j} + d_{acc,j}, \tag{8}$$

where,

$$t_{m.j} = \frac{x_{f,j} - x_1}{V_{m,j}}, \quad d_{m,j} = x_{f,j} - x_1 \tag{9}$$

where, $x_{f,j}$ denotes the vehicle's initial lane-change position and x_1 represents its post-maneuver position.

C. Optimal Trajectory Selection

Based on the preceding analysis, the trajectory-velocity profile can be obtained by combining the spatial trajectory with the trapezoidal velocity configuration. After generating multiple candidate trajectory-velocity profiles, an evaluation function is designed to select the optimal trajectory:

$$E_j = W_T T_{total,j} + W_D d_{lc} \tag{10}$$

where, W_T, W_D respectively correspond to the total trajectory duration $T_{total,j}$ and constant-velocity lane-change distance d_{lc}, emphasizing the traffic efficiency of candidate trajectories. d_{lc} is selected as a key metric because the energy consumption during constant-velocity lane-changing maneuvers is significantly higher, making it a more appropriate indicator than using $D_{sum,j}$ alone.

After calculating the evaluation metric E_j for each candidate trajectory, the trajectories are compared in terms of total time consumption and lane-change distance. In a platooning scenario, the inter-vehicle dynamics play a significant role in determining the feasibility of each trajectory. A smaller E_j value indicates a higher efficiency of the candidate trajectory for the individual vehicle in the platoon. The optimal trajectory is ultimately selected by considering both the vehicle dynamic stability indicators and the cooperative maneuvering constraints required to maintain safe, synchronized movements within the platoon.

III. Vehicle Constraints in Platoons

Determining the maximum lane-change velocity $V_{m,j}$ constitutes the core of this method, achieved by formulating an optimization problem to maximize safe lane-change speed while incorporating stability boundary constraints that account for braking-induced load transfer. This optimization framework explicitly accounts for the impact of dynamic load transfer on vehicle stability during braking, ensuring an optimal balance between driving safety and motion efficiency. The optimization problem is formulated as follows:

$$\min J_j = -V_{m,j} \tag{11}$$

$$s.t. \begin{bmatrix} \dot{V}_{y,j} \\ \dot{r}_{y,j} \end{bmatrix} = f((V_{y,j}, r_{y,j}), \delta_j) \tag{12}$$

$$V_{y,\min,j} \le V_{y,j} \le V_{y,\max,j}, \tag{13}$$

$$r_{\min,j} \le r_j \le r_{\max,j} \tag{14}$$

where, $V_{y,j}$ and r_j represent the vehicle's lateral velocity and yaw rate respectively, while δ_j denotes the steering wheel angle. The constraints $V_{y,\min,j}$ and $V_{y,\max,j}$ define the allowable bounds for lateral velocity, $r_{\min,j}$ and $r_{\max,j}$ pecify the permissible range for the yaw rate.

A. System Dynamics Constraints

Given the specific characteristics of high-speed lane-changing in truck platoons, we focus on analyzing the load-transfer effect induced by braking behavior. This effect influences the vehicle's lateral force-generating capability through tire vertical load distribution and cornering stiffness [28]. Such a coupling effect significantly alters the vehicle's

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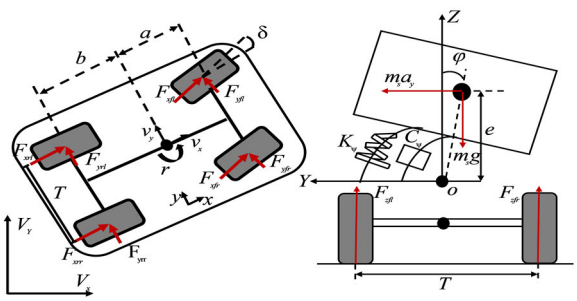

Fig. 3. Vehicle model.

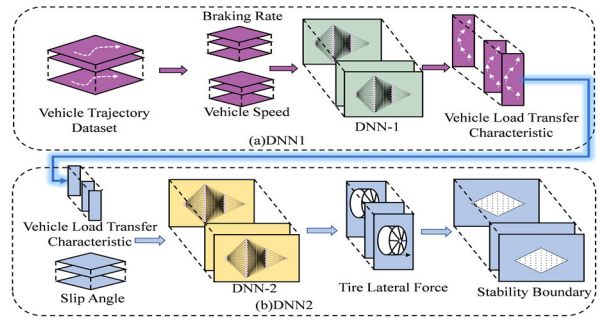

Fig. 4. Stability boundary neural network structure.

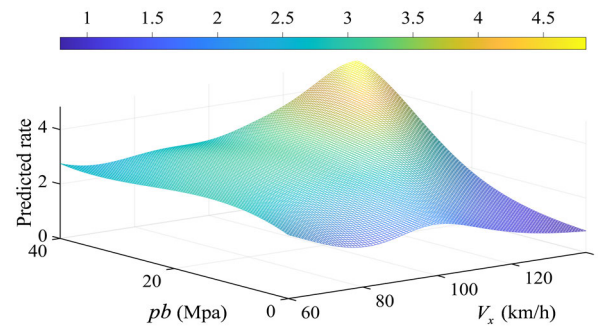

Fig. 5. Load transfer characteristics.

lateral force-generating capacity, leading to time-varying stability boundaries.

Based on vehicle dynamics analysis, this study first establishes a roll-dynamics equation that incorporates load-transfer effects. Theoretical analysis reveals that braking-induced load transfer simultaneously affects both the vehicle's lateral stability margin and energy accumulation characteristics. Similar to [14], which employed deep neural networks (DNNs) to approximate optimal parking trajectories, this paper designs a specialized neural network model for the dynamic prediction of stability boundaries. However, unlike [14], which focused on static scenario trajectory planning, the present research achieves real-time adaptive correction of stability boundaries for high-speed lane-changing scenarios by embedding load-transfer dynamics. This model effectively addresses the trajectory-feasibility misjudgment problem caused by fixed-boundary assumptions in conventional methods by updating the nonlinear relationship between tire cornering stiffness and vertical loads online. The vehicle dynamics model is shown in Figure 3. The vehicle dynamics equations are formulated as follows:

$$\begin{cases} m(\dot{V}_x - \dot{V}_y r) = (F_{xfl} + F_{xfr})\cos\delta - (F_{yfl} + F_{yfr})\sin\delta \\ \qquad\qquad + F_{xrl} + F_{xrr} \\ m(\dot{V}_y + \dot{V}_x r) = (F_{xfl} + F_{xfr})\sin\delta + (F_{yfl} + F_{yfr})\cos\delta \\ \qquad\qquad + F_{yrl} + F_{yrr} \\ I_z\dot{r} = a(F_{yfl} + F_{yfr})\cos\delta + \dfrac{T}{2}(F_{yfl} - F_{yfr})\sin\delta \\ \qquad\qquad - b(F_{yrl} + F_{yrr}) \\ + a(F_{xfl} + F_{xfr})\sin\delta + \dfrac{T}{2}(F_{xfr} - F_{xfl})\cos\delta \\ \qquad\qquad + \dfrac{T}{2}(F_{xrl} - F_{xrl}) \\ I_x\ddot{\varphi} - m_s e a_y = m_s e g\varphi - K_\psi\varphi - C_\psi\dot{\varphi} \end{cases} \tag{15}$$

where, m is the quality of the whole vehicle, m_s is the sprung mass, g is the gravitational acceleration, a and b are the distances from the center of mass of the vehicle to the front and rear axles, respectively, F_{xij} and F_{yij} represent the longitudinal and lateral forces of the four wheels, respectively, with subscripts ij: fl fr rl rr denoting the left-front, right-front, left-rear, and right-rear wheels. T is the axle track, δ is the steering wheel angle, I_x and I_z are the rolling moment of inertia and the yaw moment of inertia respectively, φ is the vehicle roll angle, C_ψ is the roll damping coefficient, K_ψ is the vehicle roll stiffness,and e is the distance from the sprung mass to the roll center, r is the vehicle yaw rate, V_x and V_y are the longitudinal and lateral velocities of the vehicle respectively, a_y is the lateral acceleration of the car body.

The load transfer induced by braking alters the load distribution and modifie a and b while simultaneously affecting the tire lateral forces. Consequently, as derived from Equation (15), r is also influenced.

B. Stability Boundary Generation Based on DNN

Regarding the prediction of vehicle stability boundaries in platoons under nonlinear tire characteristics, existing studies generally neglect the influence of dynamic load transfer [19]. The proposed DNN modules are trained and calibrated using a dataset collected from full-scale real vehicle experiments on a two-axle heavy-duty truck. The dataset is preprocessed into supervised input–output pairs for DNN1/DNN2 learning. To explicitly account for this effect, a load-transfer prediction module, DNN1, is designed, as illustrated in Fig. 4. Such a modular use of neural networks is consistent with recent DNN-based motion control research, which exploits learning to approximate specific dynamic mappings while preserving the underlying physical model structure [14], [29]. DNN1 adopts a fully connected feedforward architecture, taking the initial velocity and braking intensity as inputs and outputting the corresponding load transfer rate. As shown in Fig. 5, the simulation results indicate that the load transfer rate increases significantly with both braking intensity and vehicle speed, consistent with theoretical expectations.

Assuming identical dynamic characteristics for the front and rear wheels, the load transfer rate output by DNN1 is used

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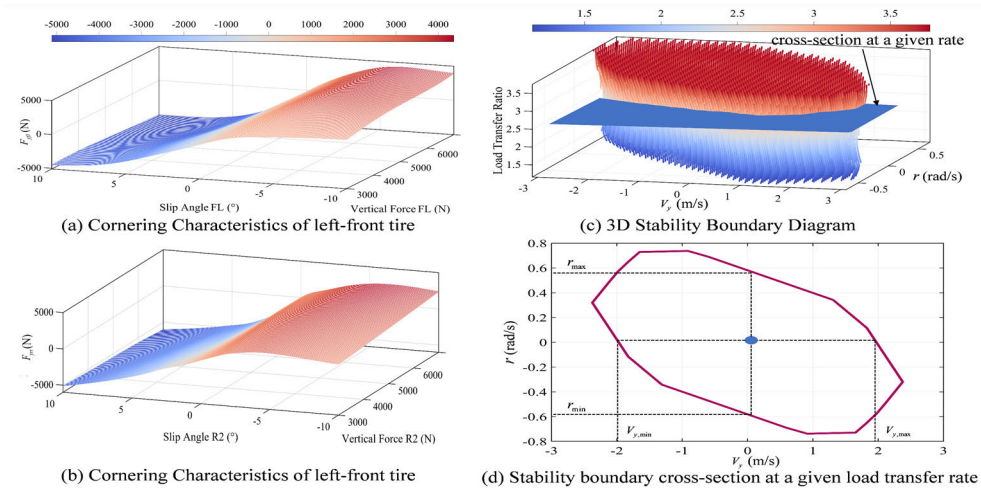

(a) Cornering Characteristics of left-front tire
(c) 3D Stability Boundary Diagram
(b) Cornering Characteristics of left-front tire
(d) Stability boundary cross-section at a given load transfer rate

Fig. 6. Tire sideslip characteristics and stability boundary.

to determine the vertical load of each wheel via axle-load distribution calculations. Figure 4(b) shows that each wheel's vertical load and slip angle inputs DNN2, which outputs the tire's lateral force. Compared with conventional methods, the neural network approach employed in this study offers significant advantages: the Magic Formula [28] characterizes tire behavior only under static conditions, whereas dynamic models such as the LuGre model [16] require linearization, which compromises dynamic features. In contrast, this data-driven method completely preserves the tire's dynamic characteristics.

Based on the assumption of identical dynamic characteristics between the front and rear wheels, the lateral force-slip angle characteristic curves for the left-front and right-rear wheels are obtained through DNN2, as shown in Figure 6(a) and (b). These curves enable accurate extraction of dynamic cornering stiffness parameters, providing critical inputs for stability boundary generation. The stability boundaries are generated using the established network by first dividing the constraints of the optimization problem (5) into two components: yaw rate r and lateral velocity V_y. Equation (15) is then simplified as follows:

$$\begin{cases} m(\dot{V}_y + V_x r) = (F_{yfl} + F_{yfr})\cos\delta + F_{yrl} + F_{yrr} \\ I_z\dot{r} = a(F_{yfl} - F_{yfr})\cos\delta + \dfrac{T}{2}(F_{yfl} - F_{yfr})\sin\delta \\ \qquad\qquad -b(F_{yrl} + F_{yrr}) \end{cases} \tag{16}$$

The lateral stability index of the vehicle is derived by applying the local linearization method from [23] to Equation (16), as follows:

$$-\frac{(a+b)\,g}{V_x^2 - (T/2)^2\,r^2} < K_{us} < \frac{2\,(a+b)\,g}{V_x^2 - (T/2)^2\,r^2}, \tag{17}$$

where,

$$K_{us} = \frac{mg\left(C_{\alpha r}b - C_{\alpha f}a\right)}{(a+b)\,C_{\alpha f}C_{\alpha r}}$$

where, K_{us} represents the lateral stability index, V_x denotes the vehicle's longitudinal velocity, $C_{\alpha f}$ is the sum of cornering stiffness for the front axle wheels, $C_{\alpha r}$ is the sum of cornering stiffness for the rear axle wheels.

This study innovatively incorporates the comprehensive influence of load transfer on vehicle dynamic parameters. Compared with [24], the proposed method accounts for both the centroid position variation caused by load transfer (making the front and rear wheelbases a and b dynamic parameters) and the nonlinear variation of tire cornering stiffness with vertical loads. The coupling of these two factors results in complex, non-monotonic stability boundaries, as shown in Figure 6(c). Figure 6(d) displays a cross-sectional view of the stability boundaries at a specific load transfer rate from Figure 6(c).

In practical applications within truck platoons, this method adopts an offline training-online lookup implementation strategy. During the offline phase, the pre-trained DNN1 establishes the mapping relationship between braking intensity and load transfer rate while precomputing various dynamic parameters corresponding to different load transfer rate values: (1) centroid position parameters, (2) vertical loads for each wheel, and (3) tire cornering stiffness obtained through the pre-trained DNN2. The restored parameter table is queried during online operation based on real-time braking conditions of both the lead and following trucks, allowing rapid determination of the current platoon-wide stability boundary. Specifically, the process can be divided into four steps:

The method converts complex neural network computations into efficient table lookups via offline preprocessing, meeting real-time requirements while maintaining accuracy. The pre-stored dynamic boundary surfaces accurately reflect stability limits under different braking intensities, ensuring safe and synchronized maneuvers within the platoon.

C. Rollover Criteria Based on Energy Accumulation

Existing rollover risk assessment methods based on real-time Load Transfer Ratio (LTR) suffer from overly conservative evaluations, primarily because they equate instantaneous wheel lift-off directly with actual rollover risk while neglecting the vehicle system's dynamic characteristics. In the context of

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truck platoons, this issue becomes more pronounced, as vehicles in tight formation experience significantly amplified load transfer effects, which can affect stability more drastically than in single-vehicle scenarios. Research demonstrates that vehicle rollover is fundamentally a dynamic process of gradual energy accumulation, influenced by a range of factors, including the system's moment of inertia and damping properties.

To address this, this paper proposes a novel rollover risk assessment method based on roll energy accumulation, establishing an energy accumulation model to more accurately characterize actual rollover risk in platoon scenarios. During the speed planning phase, this method performs online screening of candidate trajectories, effectively improving trajectory planning efficiency while ensuring driving safety.

The rollover energy threshold is the minimum energy required for a vehicle to transition from a horizontal state to the point of rollover initiation. This threshold equals the potential energy difference between the pre-rollover and rollover states, representing the minimum energy needed for the static-to-rollover transition. The potential energy difference is derived by calculating the height difference between the sprung and unsprung mass centers using the maximum roll stability angle. In this study, the rollover energy threshold calculation incorporates a load transfer correction coefficient with a value range of [0.8, 1.2], accurately reflecting the dynamic influence of load transfer on the energy threshold under varying braking intensities. The energy threshold is mathematically defined as:

$$E_t = \lambda_{et} \Delta E_p \tag{18}$$

where, E_t represents the energy threshold, λ_{et} denotes the load transfer correction coefficient, ΔE_p is the rollover potential energy difference.

In vehicle driving, the vehicle's attitude and motion determine the vehicle's energy. The attitude is related to the potential energy, and the motion is related to the kinetic energy. The rollover of the vehicle is only directly related to the lateral motion and roll motion of the vehicle. Therefore, based on the above two directions of motion, the vehicle's kinetic energy mainly includes the vehicle's lateral average kinetic energy and the sprung mass's rotational kinetic energy. The potential energy mainly consists of the potential energy due to the vehicle's gravity and the potential energy stored in the suspension deformation. The total energy is expressed as:

$$E^{lat} = E_p^{lat} + E_k^{lat} \tag{19}$$

where, E_k^{lat} represents the vehicle's lateral kinetic energy, E_p^{lat} denotes the vehicle's lateral potential energy:

$$E_k^{lat} = \frac{1}{2} m_s V_y^2 + \frac{1}{2} I_x \dot{\varphi}^2 \tag{20}$$

$$E_p^{lat} = m_s g \Delta e + \sum_{i=1}^{4} \frac{1}{2} K_i x_i^2 \tag{21}$$

where, Δe is the variation in roll center height, K_i indicates the suspension stiffness, x_i signifies the suspension deformation. with subscripts i: front, rear.

During actual vehicle operation, motions in different directions interact with each other. For instance, while longitudinal motion does not directly cause vehicle rollover, the coupling between longitudinal and lateral motions through the steering system converts translational kinetic energy during turns into rollover energy. In a platoon, this interaction is even more pronounced, as cooperative dynamics among vehicles shape how energy is distributed and accumulated across the entire convoy. Therefore, the energy stability indicator must simultaneously account for vehicle speed and wheel steering angle effects, as well as inter-vehicle coupling. Based on this analysis, we define a new vehicle rollover energy indicator as follows:

TABLE I
PROCEDURE OF STABILITY BOUNDARY GENERATION

Step	Content
1	Retrieve the load transfer rate value by querying the pre-computed table based on current vehicle speed and braking intensity;
2	Index the pre-stored parameter set $\{a, b, F_{zij}\}$ using the obtained load transfer rate;
3	Acquire the corresponding tire cornering stiffness values through F_{zij} ;
4	Substitute the dynamic parameters into boundary condition Equations (16) and (17) to generate real-time stability constraints.

$$ERI = \frac{\lambda_c E^{lat}}{E_t} + \frac{|\delta| V_x}{|\delta| V_x + \lambda_\delta} e^{\frac{-1}{V_x + \lambda_u}} \tag{22}$$

where, λ_δ denotes the speed correction coefficient for steering, λ_u indicates the speed correction coefficient λ_c signifies the inter-vehicle coupling coefficient:

$$\lambda_c = e^{-|\dot{\varphi}|}, \lambda_\delta = e$$

In Equation (22), the first term represents the energy reserve coefficient, while the second term denotes the speed-steering interference coefficient. The vehicle is most stable when the energy stability indicator equals zero; increasing values indicate greater proximity to rollover conditions. This energy metric comprehensively considers the coupled effects of vehicle speed, path geometry, road conditions, and platoon dynamics. Its deep integration into the speed planning phase enables simultaneous improvements in both trajectory safety and planning efficiency.

IV. SIMULATION RESULTS AND ANALYSIS

The trajectory planning strategy proposed in this study is validated through co-simulation, primarily focusing on lane-change maneuvers performed on high-speed dry asphalt pavement.

A. Simulation Settings

This study first establishes a co-simulation platform using Simulink and TruckSim to validate the proposed strategy. A loaded light truck is selected as the simulation object in TruckSim. The minimum snap method generates the candidate trajectory set for verification, with obstacle vehicles positioned

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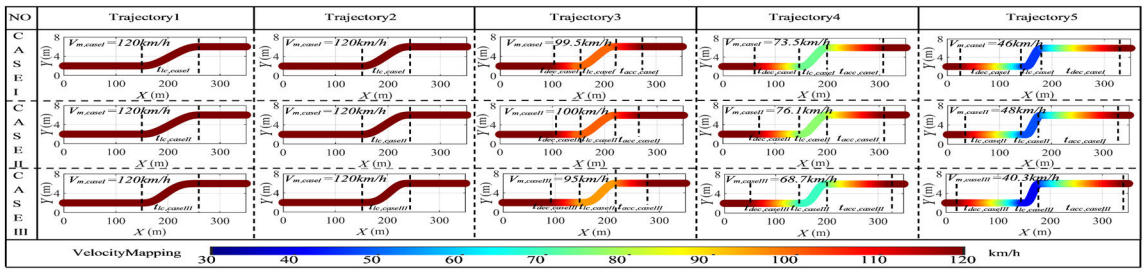

Fig. 7. Candidate trajectory-velocity profile set.

TABLE II
VEHICLE PARAMETERS IN TRUCKSIM

Parameter	Value	Parameter	Value
$Y_{start,j}$	2 m	X_{f1}	110 m
X_{f2}	90 m	X_{f3}	70 m
X_{f4}	50 m	X_{f5}	30 m
$V_{x,0}$	120 km/h	d_{safe}	150 m
a_{dec}	-4 m/s^2	a_{acc}	3 m/s^2
W_T	1 s^{-1}	W_D	0.2m^{-1}

TABLE III
SIMULATION PARAMETERS

Parameter	Value	Parameter	Value
m	9455 kg	m_s	7091 kg
g	9.8 m/s^2	e	1.175 m
I_x	4846 kg/m^2	T	1.9 m
l	5 m	I_z	73850 kg/m^2
C_ψ	36100 $N{\cdot}m{\cdot}s/rad$	K_ψ	400000 $N{\cdot}m/rad$
K_{front}	200000 N/m	K_{rear}	300000 N/m

TABLE IV
CASE I COMPARISON

Trajectory	t_{dec}/s	t_{lc}/s	t_{acc}/s	T_{sum}/s	d_{lc}/s	E
1	0	3.30	0	3.3	110	25.20
2	0	2.70	0	2.7	90	20.70
3	1.42	2.53	1.89	5.84	70	19.84
4	3.22	2.45	4.29	9.96	50	19.96
5	5.13	2.35	6.84	14.32	30	20.32

at the center of a single lane. Relevant vehicle parameters are listed in Table II. Three simulation scenarios are designed to verify the effectiveness of the proposed strategy.

Case I investigates vehicle stability characteristics when neglecting load transfer effects by fixing the load transfer rate at 1 and deriving corresponding dynamics parameters through the pre-trained network. Using Equation (17), stability boundary conditions (Equations (13)–(14)) without load transfer consideration are obtained, followed by solving the optimization problem (11) under constraints (12) to determine the maximum lane-change speed. This scenario focuses on two aspects: 1) evaluating stability performance when ignoring load transfer effects, and 2) comparing the proposed energy-based rollover indicator with conventional LTR metrics, particularly examining their sensitivity differences during high-speed lane changes to validate the new indicator's advantage in mitigating LTR's conservatism.

Case II incorporates load-transfer effects to validate the effectiveness of the proposed load-transfer compensation strategy. During simulation, the initial speed and braking intensity determined by speed planning are input to Network I to calculate the load transfer rate for each candidate trajectory. This load transfer rate is then fed into Network II to obtain corresponding dynamic parameters, establishing stability boundary conditions (Equations (13)–(14)) that account for load transfer effects through Equation (17). The final solution process remains consistent with Case I. Comparative analysis of both scenarios systematically evaluates the actual performance improvement of the proposed method in enhancing high-speed lane-change stability.

Case III employs MPC trajectory planning with static phase-plane stability boundary constraints as a comparison to verify the effectiveness of the proposed method. The simulation conditions are identical to those in Case II, with the only difference being the trajectory-planning method. In this Case, the MPC method optimizes the lane-change trajectory by incorporating static stability boundary constraints, and it is compared with the proposed method, which dynamically adjusts the stability boundaries. This setup aims to evaluate further the performance differences between the two methods under varying stability constraints and to provide a reference for subsequent analysis.

B. Case I Results

In trajectory planning without considering load transfer effects (Case I), the candidate trajectory set is generated based on stability boundary constraints (13)–(14), and the candidate trajectory set is shown in the second row of Figure 7. The X-Y plane trajectory curves, combined with velocity profiles (dark red indicating high speed and dark blue indicating low speed), demonstrate the spatiotemporal characteristics of each trajectory. Table IV data reveal that Trajectories 1 and 2, requiring no deceleration (with the maximum lane-change speed equal to initial speed), exhibit cost function values of 25.20 and 20.70, respectively, while Trajectories 3-5, requiring speed adjustment through deceleration, show sequentially increasing cost function values of 19.84, 19.96, and 20.32. If selected solely on this metric, without considering dynamic characteristics, Trajectory 3, with the minimum cost function value (19.84), would be chosen as optimal despite its significant actual dynamic stability issues.

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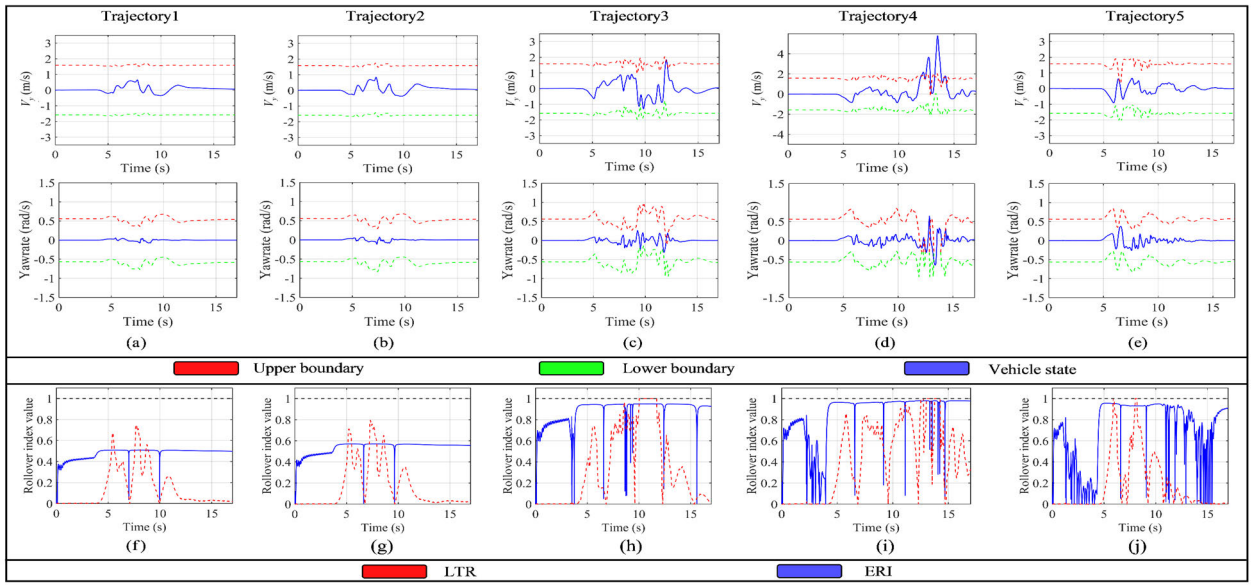

Fig. 8. Vehicle dynamic responses under different trajectories(Case I).

Analysis of the stability boundaries and rollover indicators in Figure 8 shows that Trajectories 1 and 2 strictly satisfy the boundary constraints for both lateral velocity and yaw rate; Trajectory 3 exceeds the upper limits by 3% in lateral velocity and 5% in yaw rate; Trajectory 4 violates the limits by 180% in lateral velocity and 150% in yaw rate; while Trajectory 5 approaches but does not exceed the boundaries. The peak values of both LTR and ERI indicators for Trajectories 1 and 2 remain significantly below the thresholds (ERI <0.6). In contrast, Trajectories 3-5 reach the critical LTR peak value of 1, with their ERI peak values at 92%, 99%, and 96% of the threshold, respectively. The simulation demonstrates that transient LTR violations are caused by temporary wheel lift-off (without actual rollover), while ERI more accurately characterizes the energy accumulation process.

Analysis of Case I simulation results demonstrates that selecting Trajectory 3, with the minimum cost function value (19.84), would place the vehicle in a critically unstable state due to exceeding dynamic limits. In contrast, Trajectory 4 proves completely infeasible with severe boundary violations (>150%), and Trajectory 5 - despite its slightly higher cost function value (20.32) - emerges as the optimal compromise solution in Case I by strictly satisfying all dynamic constraints, thereby revealing the inherent conflict between geometric optimization and dynamic safety in conventional trajectory planning and confirming the necessity for cooperative optimization.

C. Case II Results

In Case II, load transfer effects are considered, and the proposed co-optimization method for the dynamic stability boundary and ERI indicator demonstrates significant performance improvements. As shown in the first row of Figure 7, compared to Case I, the maximum lane-change speeds of Trajectories 3, 4, and 5 increase by 0.5%, 3.5%, and 4.3%, respectively, while the cost function values (Table V) decrease by 0.55%-2.51%. Notably, Trajectory 4 achieves the minimum cost function value of 19.46, emerging as the optimal solution with substantially superior lane-change efficiency compared to other alternatives.

TABLE V
CASE II COMPARISON

Trajectory	t_{dec}/s	t_{lc}/s	t_{acc}/s	T_{sum}/s	d_{lc}/s	E
1	0	3.3	0	3.3	110	25.20
2	0	2.7	0	2.7	90	20.70
3	1.34	2.52	1.83	5.73	70	19.73
4	3.04	2.34	4.05	9.46	50	19.46
5	5	2.26	6.67	13.92	30	19.92

Analysis of the stability indicators in Figure 9 confirms that all candidate trajectories maintain lateral velocity and yaw rate within the dynamic stability boundaries. While Trajectories 1-2 exhibit smooth dynamic responses, Trajectories 3-5, despite requiring deceleration for lane changes, fully comply with constraint conditions throughout the maneuver, validating the effectiveness of the dynamic boundary generation method. LTR and ERI metrics for Trajectories 1-2 remain below safety thresholds (ERI <0.5). Although Trajectories 3-5 transiently reach the critical LTR value of 1 (caused by momentary wheel lift-off), their ERI peaks at 94%, 98%, and 96% of the threshold, respectively, accurately characterizing the energy accumulation process without false alarms. These results demonstrate ERI's superior capability to differentiate actual rollover risks from transient disturbances, effectively overcoming the conservatism inherent in traditional LTR metrics.

Analysis of Case II simulation results confirms the engineering practicality of the proposed method, with Trajectory 4 simultaneously satisfying dynamic stability and rollover safety constraints while achieving optimal cost function performance (E = 19.46). Compared to the compromised solution of

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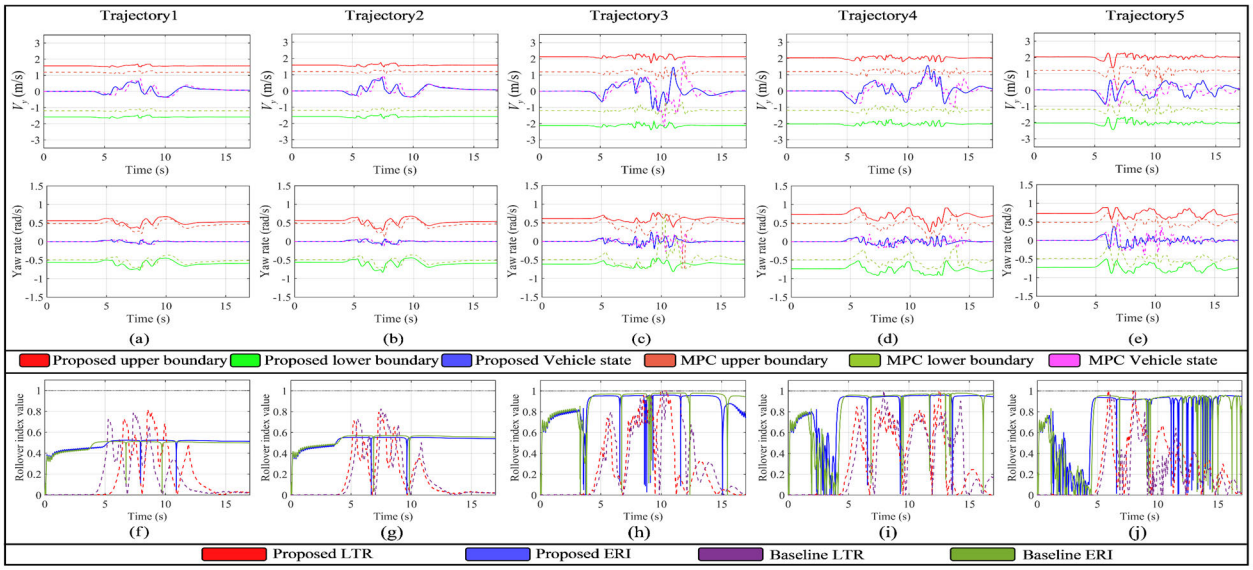

Fig. 9. Vehicle dynamic responses under different trajectories(Case II and Case III).

selecting suboptimal Trajectory 5 (E = 20.32) in Case I, co-optimizing dynamic stability boundaries and ERI indicators significantly enhances the comprehensive performance of the planning system.

D. Case III Results

In the simulation results, the trajectory-velocity candidate set generated by the MPC baseline scheme based on static boundaries is presented in the third row of Fig. 7. Compared with the proposed dynamic-boundary cooperative framework, the maximum lane-change velocities for Trajectory 1 and Trajectory 2 are nearly identical under both approaches. In contrast, for Trajectory 3, Trajectory 4, and Trajectory 5, the maximum lane-change velocities achieved by the proposed method are approximately 5.3%, 15.8%, and 19.1% higher, respectively, than those permitted under the baseline scheme without stability violation. Stability analysis indicates that the state trajectories of Trajectory 1 and Trajectory 2 remain entirely within their respective stability boundaries under both methods, as shown in Fig. 9(a) and (b). However, for Trajectory 3, Fig. 9(c) illustrates that the pink state curve under the baseline scheme would significantly exceed the static phase-plane boundary at the speed planned by the proposed method, whereas the blue one remains stable within the dynamic boundary framework. As depicted in Fig. 9(d), even at a lower speed, the pink state curve of Trajectory 4 under the baseline approach closely approaches the static boundary. For Trajectory 5, the pink state variable r under the baseline scheme exceeds the static boundary by approximately 5%, as seen in Fig. 9(e). In comparison, under the dynamic boundary scheme, both Trajectory 4 and Trajectory 5 maintain considerable safety margins relative to their respective boundaries at the planned speeds.

Regarding rollover indicators, the peak values of both LTR and ERI for Trajectory 1 and Trajectory 2 remain well below their respective thresholds under both methods. In contrast, Trajectory 3, Trajectory 4, and Trajectory 5 all reach the LTR threshold; however, simulations confirm that these represent only transient critical rollover events rather than complete rollovers. Correspondingly, under the baseline scheme, the peak ERI values for these three trajectories are approximately 93%, 98%, and 95%, respectively, while under the dynamic boundary framework, the ERI values approach but do not exceed the threshold. This demonstrates ERI's superior discriminative accuracy in identifying critical operating conditions.

TABLE VI
CASE III COMPARISON

Trajectory	t_{dec}/s	t_{lc}/s	t_{acc}/s	T_{sum}/s	d_{lc}/s	E
1	0	3.3	0	3.3	110	25.20
2	0	2.7	0	2.7	90	20.70
3	1.725	2.65	2.3	6.675	70	20.675
4	3.76	2.74	5.02	10.92	50	20.92
5	5.53	2.68	7.37	15.58	30	21.58

Finally, according to the comprehensive evaluation metrics in Table VI, Trajectory 3 under the baseline scheme achieves the minimum cost value of 20.675. However, due to its instability, it is excluded, and Trajectory 2, with a cost of 20.7, is selected as the optimal trajectory. In contrast, the dynamic boundary cooperative framework selects Trajectory 3, with a lower cost of 19.46, as the optimal solution. The comprehensive evaluation metric of the optimal trajectory under the baseline scheme is approximately 6.4% higher than that under the proposed framework. This quantitative outcome further demonstrates that the proposed framework not only avoids overly conservative instability judgments for controllable critical conditions but also improves lane-change efficiency and more reasonably allocates stability margins while ensuring stability and rollover safety.

V. SIMULATION RESULTS DISCUSSION

This study highlights the importance of cooperative optimization of dynamic constraints in trajectory planning for

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truck platoons by comparing the proposed dynamic stability boundary method that considers load transfer effects (Case II) with conventional approaches (Case I). Experimental results show that adopting the proposed method increases the maximum lane-change speeds of candidate Trajectories 3, 4, and 5 by 0.5%, 3.5%, and 4.3%, respectively, while strictly maintaining lateral velocity and yaw rate within the dynamic stability boundaries of the entire platoon. In particular, Trajectory 3 achieves improved dynamic stability at 100 km/h in Case II compared to 99.5 km/h in Case I, confirming the nonlinear coupling between speed planning and dynamic responses. Furthermore, when compared against the MPC benchmark with static phase-plane boundaries (Case III), the proposed method demonstrates a more balanced performance. Although the nominal speeds of Trajectories 3-5 in Case III are lower than those achieved in Case II, the static-boundary approach results in trajectories that either violate stability constraints (Trajectory 3) or operate with critically low safety margins (Trajectories 4/5). This contrast underscores that the proposed dynamic boundaries enable higher, feasible speeds while robustly guaranteeing stability.

In rollover risk assessment, the proposed ERI indicator demonstrates significant advantages for cooperative platooning scenarios. When the LTR metric reaches its critical value of 1, the corresponding ERI peaks for Trajectories 3, 4, and 5 in Case I are 92%, 99%, and 96% of the threshold, respectively, whereas in Case II, they improve to 94%, 98%, and 96%. A parallel assessment under the Case III framework yields ERI peaks of approximately 93%, 98%, and 95% for the same trajectories, consistently below the threshold despite triggering LTR alarms. This consistent behavior of ERI across both Case II and Case III confirms its reliability in distinguishing critical energy accumulation from transient disturbances, irrespective of the underlying stability boundary model. These results confirm that the ERI-based energy accumulation analysis more accurately distinguishes genuine rollover risks from transient disturbances, such as brief wheel lift-off events, effectively reducing the conservative false alarm rate to below 2% compared to LTR-based methods, particularly in tight platoon formations.

Furthermore, co-simulation validation demonstrates that the proposed coordinated optimization strategy achieves a superior trade-off between safety and efficiency. In Case II, Trajectory 4 satisfies all dynamic safety constraints while attaining the minimum cost function value of 19.46, representing a 4.2% performance gain over the suboptimal solution in Case I. This advantage is further quantified against the MPC benchmark (Case III). The static-boundary constraints in Case III incorrectly exclude higher-performance trajectories due to overly conservative stability judgments, forcing the selection of a suboptimal trajectory with a cost value of 20.7. Consequently, the comprehensive cost of the chosen trajectory in Case III is approximately 6.4% higher than the optimal value achieved in Case II. This result conclusively demonstrates that the proposed framework uniquely identifies and enables more efficient trajectories, which are erroneously filtered out by conventional methods, while rigorously guaranteeing dynamic stability.

VI. Conclusion

This paper presents an adaptive trajectory planning framework for autonomous heavy-duty truck platoons that explicitly accounts for the nonlinear effects of braking-induced load transfer on lateral stability. The proposed method forms a learning-enabled, constraint-aware motion planning process by integrating a physics-informed neural architecture for online prediction of the stability boundary with an energy-based rollover index (ERI) that captures cumulative dynamic risk. In the context of platoon operations, the framework ensures that all platoon vehicles, both lead and following trucks, perform coordinated maneuvers while maintaining safe, stable conditions during high-speed lane changes.

Simulation results show that the proposed framework increases maximum feasible lane-change speeds by 0.5% to 4.3% while ensuring that critical safety indicators—including lateral velocity and yaw rate—remain within dynamically updated limits for all vehicles in the platoon. The ERI metric significantly enhances rollover risk discrimination, reducing false alarm rates to below 2% compared to conventional LTR-based criteria, particularly in tight platoon configurations. Co-simulation using MATLAB/Simulink and TruckSim confirms that the coordinated optimization strategy yields a 4.2% improvement in overall planning performance while maintaining high levels of dynamic safety.

Future research will shift focus towards evaluating and enhancing the robustness of the proposed framework under real-world V2V platooning conditions, with particular emphasis on the impact of communication delays, packet loss, and other network-induced disturbances. Corresponding mitigation strategies, such as robust predictive compensation, distributed consensus under latency constraints, and resilient trajectory coordination protocols, will be investigated to maintain stability and safety in dynamic, uncertain communication environments. In this regard, robust MPC architectures with dual-loop disturbance observers, which have been successfully applied to autonomous hypersonic vehicles, provide a valuable reference for handling strong nonlinearities and modeling uncertainties in safety-critical maneuvers [30].

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